

Exercise 4 – Number System (Binary Coded Number)

Octal to Binary As octal numbers digits range from 0 to 7, they can be represented by three binary digits.	Hexa-Decimal to Binary Similarly hexa-decimal digits and symbols can be represented using 4 binary digits (or a nibble).
0 -----> 000 1 -----> 001 2 -----> 010 3 -----> 011 4 -----> 100 5 -----> 101 6 -----> 110 7 -----> 111	0 -----> 0000 1 -----> 0001 2 -----> 0010 3 -----> 0011 4 -----> 0100 5 -----> 0101 6 -----> 0110 7 -----> 0111 8 -----> 1000 9 -----> 1001 A -----> 1010 B -----> 1011 C -----> 1100 D -----> 1101 E -----> 1110 F -----> 1111
With above, any number from octal number system can be represented using above coding e.g. $(1567)_8 = (001\ 101\ 110\ 111)_2$ $(457)_8 = (100\ 101\ 111)_2$ $(1001)_8 = (001\ 000\ 000\ 001)_2$	With above, any number from hexa-decimal number system can be represented using above coding e.g. $(45A)_8 = (0100\ 0101\ 1010)_2$ $(F48)_8 = (1111\ 0100\ 1000)_2$ $(103)_8 = (0001\ 0000\ 0011)_2$
Similarly, any binary number can be translated to an octal number by simply grouping 3bits from least significant digit to the most significant digit e.g., $(11110000011)_2 = (11\ 110\ 000\ 011)_2 = (3603)_8$ $(1101101010)_2 = (1\ 101\ 101\ 010)_2 = (1552)_8$ $(110011)_2 = (110\ 011)_2 = (63)_8$	Similarly, any binary number can be translated to a hexa-decimal number by simply grouping 4bits from least significant digit to the most significant digit e.g., $(11110000011)_2 = (111\ 1000\ 0011)_2 = (383)_{16}$ $(1101101010)_2 = (11\ 0110\ 1010)_2 = (36A)_{16}$ $(110011)_2 = (11\ 0011)_2 = (33)_{16}$

Question: Convert following numbers using binary coded number notation:

$(1111110010100110)_2 = (?)_8$	$(10111100001)_2 = (?)_{16}$	$(111100001101011)_2 = (?)_{16}$	
$(11010101111)_2 = (?)_8$	$(100001110)_2 = (?)_{16}$	$(111111110)_2 = (?)_{16}$	
$(11110010101111)_2 = (?)_8$	$(10101011101)_2 = (?)_{16}$	$(111111111111111)_2 = (?)_{16}$	
$(65123)_8 = (?)_2$	$(345)_2 = (?)_2$	$(1142)_8 = (?)_2$	$(7631)_8 = (?)_2$
$(A41C)_{16} = (?)_2$	$(10EF)_{16} = (?)_2$	$(CDE)_{16} = (?)_2$	$(1000)_{16} = (?)_2$
$(1C8)_{16} = (?)_8$	$(657)_8 = (?)_{16}$	$(14F)_{16} = (?)_8$	$(DEF)_{16} = (?)_8$